02/03

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=) Leetcode Problems on GCD.

1) GCD of odd & even sums

constraints : $1 <= n <= 1000$

```
int gcd(int a, int b){
  while (b){
    int t = a%b;
    a = b;
    b = t;
  }
  return a;
}
int gcdOfOddEvenSums (int n){
  int odd = n*n;
  int even = n*(n+1);
  return gcd (odd, even);
}
```

2) Sum of GCD of Formed pairs

Constraints: $1 <= n == nums.length <= 10^5$

$1 <= nums[i] <= 10^9$

```
int gcd(int a, int b){
  while (b){
      int t= a%b;
      a= b;
      b= t;
  }
  return a;
}

int sumOfGCD (int *nums, int numSize){
  int sum=0;
  for (int i=0; i<numSize; i++){
      for (int j=i+1; j<numSize; j++){
          sum += gcd(nums[i], nums[j]);
      }
  }
  return sum;
}
```

3) Number of Subarrays with GCD equal to k.

Constraints: $1 <= nums.length <= 1000$
$1 <= nums[i], k <= 10^9$

```
int gcd (int a, int b){
  while (b){
    int t = a%b;
    a = b;
    b = t;
  }
  return a;
}

int subarrayGCD (int* nums, int numSize, int k){
  int count = 0;
  for (int i=0; i< numsSize; i++){
    int g = 0;
    for (int j = i; j <numsSize; j++){
      g = gcd (g, nums[j]);
      if (g==k) count++;
      if (g<k) break;
    }
  }
  return count;
}
```

Output:

4) Minimum split into subarrays with gcd > 1

constraints: $1 <= nums.length <= 10^5$
$1 <= nums[i] <= 10^7$

```
int gcd (int a, int b){
  while (b){
    int t = a%b;
    a = b;
    b = t;
  }
  return a;
}

int minimumSplits(int *nums, int numsSize){
  int g = 0, ans = 1;
  for(int i = 0; i < numsSize; i++){
    g = gcd (g, nums[i]);
    if (g == 1){
      ans++;
      g = nums[i];
    }
}
```

```
    return ans;
}
```

output :

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